

Supplementary Information for:

Constructing the Nation: Identity and Historical
Narratives in South Korean History Textbooks

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Appendix A: Korean identity terms and term-discovery robustness

Appendix A.1: Glossary of Korean terms

Throughout the paper each Korean identity concept is given by its English translation in italics, with the McCune–Reischauer romanization in parentheses at first mention (for example, *nation* (minjok)). Thereafter the English term is used on its own. Institutional terms that have no concise English equivalent are introduced by an English gloss once and then carried in romanization (for example, *kukchǒngje*). This glossary lists every such term with its romanization and Hangul. Proper nouns (dynasties, persons, places, publishers, and book titles) are given in romanization in the running text and are not repeated here.

English (in text)	McCune–Reischauer	Hangul
<i>nation</i>	minjok	민족
<i>state</i>	kukka	국가
<i>citizens</i>	kungmin	국민
<i>independence</i>	tongnip	독립
<i>democracy</i>	minjujuŭi	민주주의
<i>unification</i>	t’ongil	통일
<i>liberation</i>	haebang	해방
<i>history</i>	yŏksa	역사
<i>territory</i>	yŏngt’o	영토

Table S1: Core identity concepts (the nine analyzed terms).

English (in text)	McCune–Reischauer	Hangul	Note
single-textbook, state-authored system	kukchǒngje	국정제	state monopoly authorship, 1974–2002; one official textbook
certified multi-publisher system	kǒmjǒngje	검정제	certification system restored in 2002; competing publishers
national history (subject)	kuksa	국사	the school subject and textbook genre analyzed
social studies (subject)	sahoe	사회	a more revision-flexible subject; contrast to <i>kuksa</i>
modern & contemporary Korean history	Han’guk kŭnhyŏndaesa	한국근현대사	post-2002 high-school textbook genre/series
National Institute of Korean History (NIKH)	Kuksa P’yŏnch’an Wiwŏnhoe	국사편찬위원회	source archive for most of the corpus
ministry textbook certification (Japan)	kentei	検定	Japanese Ministry of Education screening (comparative reference)

Table S2: Institutional and system terms.

English (in text)	McCune–Reischauer	Hangul
national community	minjok kongdongch’e	민족 공동체
great national unity	minjok taedan’gyŏl	민족 대단결
national identity	minjok chŏngch’esŏng	민족 정체성
our nation	uri minjok	우리 민족
the Korean people	hanminjok	한민족
compatriots	tongp’o	동포
homogeneous (single) nation	tanil minjok	단일민족
multiethnic	taminjok	다민족
multicultural	tamunhwa	다문화
citizenship	siminkwŏn	시민권
migrants	ijumin	이주민
foreigners	oegugin	외국인
world citizen	segye simin	세계 시민

Table S3: Descriptive phrases and framing markers.

Term-discovery robustness

This appendix reports three robustness checks on the term-discovery pipeline. The first examines whether the choice of model scale within the EXAONE family affects which terms are selected. The second tests whether the final nine terms are stable across four 32B-class language models from independent model families (a cross-family ensemble designed to catch model-specific extraction artifacts). The third checks whether BERTopic and LDA, two conventional topic-modeling methods, independently recover the same terms from the corpus. The three checks address different threats to the validity of the discovery procedure, namely within-family scale sensitivity, cross-family generalizability, and recovery by methods that impose no prior on identity relevance.

Model-scale sensitivity within EXAONE

We first ran the extraction pipeline at three parameter scales using models from the same EXAONE family: EXAONE 3.5 at 2.4B and 7.8B parameters, and EXAONE-Deep at 32B (4-bit quantized). All other pipeline settings were held constant: the same Korean zero-shot prompt, the same adaptive window sampling, the same scoring formula, and the same $wf \geq 50$ threshold per model.

Table S4 reports the results. All three scales achieve perfect recall. Every model selects all nine final terms. Precision varies substantially. The 2.4B model selects 37 terms, of which only 9 (24%) are in the final set. The remaining 28 include generic political and social vocabulary (정치, 사회, 문화, 가치, 외교) alongside historical proper nouns and low-salience items. At 7.8B precision jumps to 60% (9 hits from 15 terms). The 32B model selects 16 terms at 56% precision, comparable to 7.8B, but the six additional terms differ in kind. Where the 7.8B model’s extras are conceptually ambiguous (정치, 사회, 공동체, 문화, 가치, 외교), the 32B model’s extras are predominantly proper nouns (한국, 한국인, 조선, 고려, 고구려) that would be excluded from the final set on theoretical grounds regardless of precision. Restricting to conceptual rather than referential terms, the 32B model selects only two items outside the consensus set (우리, 사람), yielding an effective conceptual precision of 82%. The 2.4B model produces more conceptual false positives, which is consistent with the smaller model being less able to discriminate identity-salient concepts from generic historical vocabulary.

Model size	Terms selected	Hits on 9	Precision	Recall	Missed
2.4B	37	9	24%	100%	none
7.8B	15	9	60%	100%	none
32B	16	9	56%	100%	none

Table S4: EXAONE extraction performance by model scale.

Note: Precision is the share of selected terms in the final nine-term set. All three scales achieve perfect recall. The 32B model’s six additional selections are predominantly proper nouns excluded on theoretical grounds; its effective conceptual precision is 82%.

The scale comparison motivates using the 32B model for the main ensemble. Scale improvement plateaus after 7.8B in raw precision, but the 32B model is qualitatively better behaved. It identifies the same nine terms while adding only proper nouns and pronouns rather than conceptually ambiguous items.

Cross-family ensemble: four 32B-class models

The second check asks whether the nine terms survive extraction by models from entirely different training traditions. We ran the same pipeline with four 32B-class models: EXAONE-Deep (LG AI Research, Korea), Aya Expanse (Cohere, Canada), Qwen2.5 (Alibaba, China), and Gemma 3 (Google, United States). The models differ in training corpus composition, pretraining objectives, and the language communities they were developed to serve. None shares training data or architecture with the others in any direct way, which makes cross-family agreement a conservative test of whether the extracted terms reflect real structure in the Korean corpus rather than shared model-family biases.

Each model received the same Korean zero-shot prompt, the same adaptive window sampling, and the same $wf \geq 50$ absolute floor. We replaced the earlier $\mu + 2\sigma$ threshold because it is sensitive to high-frequency demonym terms that inflate the distribution in some model runs. An absolute floor yields more comparable cut-offs across models that differ in extraction volume. A term entered the final set if at least three of the four models selected it. We additionally excluded proper nouns (terms denoting specific historical entities such as kingdoms, dynasties, and the country itself) on the grounds that they function as referential labels rather than contested conceptual constructs. The semantic-axis method used in the main analysis is designed to track how the meaning of a concept shifts across political periods. Referential terms do not have a comparable axis of contested meaning and would not be interpretable in that framework.

Table S5 reports the voting outcome. Nine terms are selected by at least three of the four models and pass the proper noun filter: 민족, 국가, 국민, 독립, 민주주의, and 통일 at 4/4

consensus; 해방, 역사, and 영토 at 3/4. Five additional terms reach 4/4 consensus but are excluded as proper nouns (한국, 한국인, 조선, 고려, 고구려); three further terms score 3/4 and are excluded as proper nouns or as non-conceptual items (백제, 신라, 삼국 as dynastic kingdoms; 우리 as a first-person pronoun; 근대 as a temporal-periodization label). Three conceptual terms that appeared in the earlier EXAONE-only analysis, 정치 (politics), 공동체 (community), and 사회 (society), reach only 2/4 consensus in the cross-family ensemble and are excluded.

Term	Roman.	Gloss	EX	AY	QW	GM	Vote	Status
민족	<i>minjok</i>	ethnic nation	✓	✓	✓	✓	4/4	Final
국가	<i>kukka</i>	state/nation	✓	✓	✓	✓	4/4	Final
국민	<i>kungmin</i>	citizenry	✓	✓	✓	✓	4/4	Final
독립	<i>tongnip</i>	independence	✓	✓	✓	✓	4/4	Final
민주주의	<i>minjujuui</i>	democracy	✓	✓	✓	✓	4/4	Final
통일	<i>t'ongil</i>	unification	✓	✓	✓	✓	4/4	Final
해방	<i>haebang</i>	liberation	✓	✓	✓	—	3/4	Final
역사	<i>yŏksa</i>	history	✓	✓	✓	—	3/4	Final
영토	<i>yŏngt'o</i>	territory	✓	✓	—	✓	3/4	Final
한국	<i>han'guk</i>	Korea	✓	✓	✓	✓	4/4	Proper noun
한국인	<i>han'gugin</i>	Korean (person)	✓	✓	✓	✓	4/4	Proper noun
조선	<i>Chosŏn</i>	Chosŏn/ Korea	✓	✓	✓	✓	4/4	Proper noun
고려	<i>Koryŏ</i>	Koryŏ dynasty	✓	✓	✓	✓	4/4	Proper noun
고구려	<i>Koguryŏ</i>	Koguryŏ kingdom	✓	✓	✓	✓	4/4	Proper noun
정치	<i>chŏngch'i</i>	politics	—	✓	✓	—	2/4	Below threshold
공동체	<i>kongdongch'e</i>	community	—	✓	✓	—	2/4	Below threshold
사회	<i>sahoe</i>	society	—	✓	✓	—	2/4	Below threshold

Table S5: Four-model ensemble voting.

Note: EX = EXAONE-Deep 32B; AY = Aya Expanse 32B; QW = Qwen2.5 32B; GM = Gemma 3 27B. The top panel shows the nine final terms. The middle panel shows terms excluded as proper nouns despite high consensus. The bottom panel shows selected conceptual terms selected by fewer than three of the four models.

Table S6 compares extraction performance across models using the nine final terms as

the reference set. EXAONE-Deep achieves the highest precision (56%) and perfect recall, selecting the fewest total terms while missing none of the nine. The multilingual and non-Korean-primary models select substantially more terms (between 30 and 52) with precision ranging from 15% to 30%. Gemma 3, which is the most English-primary model in the ensemble, misses two terms (역사와 해방); Qwen2.5 misses one (영토).

The precision gap between EXAONE and the other models is consistent with the Korean-primary model having finer discrimination between identity-salient vocabulary and generic historical nouns. Examining each model’s unique selections (terms it chose that no other model selected) makes the pattern clearer. EXAONE’s single unique selection is 사람 (person, 7,150 corpus occurrences), a term that could plausibly enter an identity-extraction output. Qwen2.5’s 14 unique selections are dominated by generic historical items: 운동 (movement, 14,009 occurrences), 전쟁 (war, 4,868), 지방 (province, 5,216), 귀족 (aristocracy, 2,474), and 과거 (past/civil-service examination, 1,288). These words appear frequently in Korean history text for reasons having nothing to do with national identity. The Korean-trained model does not select them.

Model	Origin	Selected	Hits on 9	Precision	Recall	Missed
EXAONE-Deep 32B	Korean	16	9	56%	100%	none
Aya Expanse 32B	Canadian (multilingual)	30	9	30%	100%	none
Gemma 3 27B	US (English-primary)	32	7	22%	78%	역사, 해방
Qwen2.5 32B	Chinese (multilingual)	52	8	15%	89%	영토

Table S6: Extraction performance for each model in the four-family ensemble.

Note: Precision and recall are computed against the nine final terms. EXAONE-Deep achieves the highest precision and perfect recall; the multilingual and English-primary models over-extract substantially.

Independent validation: BERTopic and LDA

The third check asks whether the nine terms surface in the corpus when it is organized by methods that impose no prior on national identity. We ran BERTopic (Grootendorst, 2022) and Latent Dirichlet Allocation (LDA; Blei et al., 2003) using the same specifications as the earlier EXAONE-only analysis.

For BERTopic, each textbook was split into paragraph-scale chunks targeting roughly 500 characters.¹ This yielded 29,568 chunks with a mean length of 433.34 characters. Chunks were embedded with `BM-K/KoSimCSE-roberta-multitask` (the same model used in the main semantic analysis; Kim, 2021), then projected with UMAP and clustered with HDBSCAN. Topic representations used c-TF-IDF with a KiwiPiePy noun tokenizer. With a minimum topic size of 30, BERTopic identified 214 non-outlier topics, with 8,804 chunks (29.8%) classified as outliers. A term counted as recovered if it appeared among the top 30 representative words of at least one topic.

For LDA, we used the noun-extracted `processed_text` field as a document-level bag-of-words representation of the 67 textbooks. After filtering rare and very common terms, the vocabulary contained 20,155 tokens. We estimated models for $k \in \{10, 15, 20, 25, 30\}$ and selected by c_v coherence with 20 passes. The best model was $k = 20$ with $c_v = 0.5164$. Recovery was defined identically to BERTopic.

¹Paragraph-scale segmentation keeps inputs within the context window of the embedding model and ensures each chunk carries locally coherent content.

Term	BERTopic	LDA	Both
민족 (<i>minjok</i> , ethnic nation)	Yes (2)	Yes (1)	Yes
국가 (<i>kukka</i> , state)	Yes (22)	No	No
국민 (<i>kungmin</i> , citizenry)	Yes (1)	Yes (3)	Yes
독립 (<i>tongnip</i> , independence)	Yes (1)	Yes (3)	Yes
민주주의 (<i>minjujuui</i> , democracy)	Yes (3)	Yes (27)	Yes
통일 (<i>t'ongil</i> , unification)	Yes (3)	Yes (10)	Yes
해방 (<i>haebang</i> , liberation)	Yes (9)	No	No
역사 (<i>yŏksa</i> , history)	Yes (1)	No	No
영토 (<i>yŏngt'o</i> , territory)	Yes (4)	No	No
Coverage	9/9 (100%)	5/9 (56%)	5/9 (56%)

Table S7: Recovery of the nine identity terms by BERTopic and LDA.

Note: Parentheses report the best rank achieved across topics in BERTopic. LDA coverage figures reflect the best $k = 20$ specification.

BERTopic recovers all nine terms (100%). LDA recovers five of nine (56%). The four LDA misses are interpretable. 국가 (state) and 역사 (history) appear across virtually the entire corpus (in 64 and 67 textbooks respectively). LDA rewards terms that discriminate one topic from another. Terms present in almost every document carry little discriminative weight and rarely rank highly within any topic. 해방 (liberation, 46 textbooks, 507 occurrences) and 영토 (territory, 60 textbooks, 1,181 occurrences) fall between the two extremes, wide enough to be depressed in topic rankings yet concentrated enough not to surface reliably. BERTopic operates on 29,568 local chunks rather than 67 full documents. Finer resolution and semantic rather than co-occurrence clustering allow it to recover all four terms LDA misses.

The LDA gap is informative. The five terms LDA recovers (민족, 국민, 독립, 민주주의, 통일) are the most topically discriminative identity concepts in the corpus. LDA finds them precisely because they cluster around specific historical periods and events. The four it misses are more diffusely distributed across the entire corpus, which is itself substantively meaningful. State, liberation, history, and territory are present across all periods rather than concentrated in specific events.

Discussion

The three robustness checks provide convergent support for the nine-term set through independent procedures. Within the EXAONE family, precision rises steeply from 2.4B to 7.8B and stabilizes at 32B. All three scales achieve perfect recall. The recall convergence itself carries weight. Open-weight Korean-trained models at three substantially different capacities all select the same nine terms, which suggests the term set is a property of the corpus rather than of the largest model alone. Across model families, six of the nine terms are confirmed by all four 32B-class models and the remaining three by three of four. The Korean-trained model extracts the nine terms with markedly higher precision than multilingual or English-primary alternatives, consistent with its greater ability to distinguish identity-salient vocabulary from general-purpose historical nouns in Korean text. BERTopic independently recovers all nine terms. LDA recovers five, with the four misses attributable to known properties of document-level bag-of-words modeling.

Each procedure answers a different question. Scale sensitivity within EXAONE tests whether the discovery is an artifact of a particular parameter regime. Cross-family diversity tests whether it is an artifact of a particular training tradition. BERTopic and LDA test whether independent corpus-organization methods converge on the same concepts. The three checks provide a reasonable basis for treating the nine terms as reliable signals from the corpus.

The main text notes that three conceptual terms prominent in earlier EXAONE-only analyses (정치, 공동체, and 사회) reach only 2/4 consensus in the cross-family ensemble and are excluded. The cross-family ensemble applies a more demanding criterion than within-family multi-scale agreement. Excluding terms selected by fewer than three of the four models tightens the final set without cost, since these three terms carry no independent analytical weight in the findings.

Appendix B: Corpus inventory

Table S8 lists all 67 textbooks in the merged corpus, grouped by the five-period classification used in the main text. The table follows the metadata stored in `nikh_corpus_merged.csv`. We include each textbook’s stable identifier (`book_id`) so the printed inventory can be matched directly to the replication files. The table reproduces the fields stored in the merged corpus rather than inferring missing values. For five of the earliest items the original publisher is not recorded and the cell is left blank.

Period	book_id	Title	Year	Publisher	Lv.
Late Chosŏn	ta_p101r	대동청사 1책	1895	개인출판	Gen.
	ta_p102r	대동청사 2책	1895	개인출판	Gen.
	ta_p111r	국조사	1895	개인출판	Gen.
	ta_p11r	조선역사 상	1895	학부 편집국	Pub.
	ta_p12r	조선역사 중	1895	학부 편집국	Pub.
	ta_p13r	조선역사 하	1895	학부 편집국	Pub.
	ta_p21r	조선역대사략 권1	1895	학부 편집국	Pub.
	ta_p22r	조선역대사략 권2	1895	학부 편집국	Pub.
	ta_p23r	조선역대사략 권3	1895	학부 편집국	Pub.
	ta_p71r	보통교과 동국역사(1권)	1899		E
	ta_p72r	보통교과 동국역사(2권)	1899		E
	ta_p81r	동국사략 상	1906		Gen.
	ta_p82r	동국사략 하	1906		Gen.
	ta_p91r	초등대한역사	1908		E
Colonial	ta_p31r	심상소학국사보충아동용 - 1	1940	문부성	E
	ta_p32r	심상소학국사보충아동용 - 2	1940	문부성	E
	ta_p41r	심상소학국사보충교재 교수참고서 - 1	1940	문부성	E
	ta_p42r	심상소학국사보충교재 교수참고서 - 2	1940	문부성	E
Postwar	ta_p51r	국사교본	1946	진단학회	Gen.
	ta_p61r	우리나라의 생활	1948	진단학회	Gen.
	ta_e11	초등학교 사회생활 6-1(1차)	1954	문교부	E
	ta_h11	고등학교 국사 1차	1954	문교부	H
	ta_m11	중학교 국사 1차	1954	문교부	M
	ta_e21	초등학교 사회 6-1(2차)	1963	문교부	E
	ta_h21	고등학교 국사 2차	1963	문교부	H
	ta_m21	중학교 국사 2차	1963	문교부	M
Authoritarian	ta_e31	초등학교 국사 5(3차)	1973	문교부	E
	ta_e32	초등학교 국사 6(3차)	1973	문교부	E
	ta_h31	고등학교 국사 3차	1973	문교부	H
	ta_h32	고등학교 세계사(3차)	1973	문교부	H
	ta_m31	중학교 국사 3차	1973	문교부	M
	ta_e41	초등학교 사회 5-2(4차)	1981	문교부	E
	ta_e42	초등학교 사회 6-2(4차)	1981	문교부	E
	ta_h41	고등학교 국사 4차(상)	1981	문교부	H
	ta_h42	고등학교 국사 4차(하)	1981	문교부	H
	ta_m41	중학교 국사 4차(상)	1981	문교부	M
	ta_m42	중학교 국사 4차(하)	1981	문교부	M
	ta_e51	역대 국사교과서	1987	교육부	E
	ta_e52	초등학교 사회 6-1(5차)	1987	교육부	E
	ta_h51	고등학교 국사 5차(상)	1987	교육부	H
	ta_h52	고등학교 국사 5차(하)	1987	교육부	H
	ta_m51	중학교 국사 5차(상)	1987	교육부	M
	ta_m52	중학교 국사 5차(하)	1987	교육부	M

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Table S8 continued

Period	book_id	Title	Year	Publisher	Lv.
	H-18(1,90)1	고등학교 국사 (상)	1990	문교부	H
	H-19(1,90)1	국사	1991	문교부	M
Democratic	ta_e61	초등학교 사회 6-1(6차)	1995	교육부	E
	ta_h61	고등학교 국사 6차(상)	1995	교육부	H
	ta_h62	고등학교 국사 6차(하)	1995	교육부	H
	ta_m61	중학교 국사 6차(상)	1995	교육부	M
	ta_m62	중학교 국사 6차(하)	1995	교육부	M
	H-21(1,2001)1	국사 편찬 위원회	2001	문교부	M
	H-2(1,2002)	국사 편찬 위원회	2002	교육인적자원부	H
	H-6(1,2007)	한국근현대사	2002	중앙교육진흥연구소	H
	H-7(1,2007)	한국근현대사	2002	법문사	H
	H-9(5,2007)	한국근현대사	2002	대한교과서(주)	H
	ta_e71	초등학교 사회 6-1(7차)	2002	교육인적자원부	E
	ta_h71	고등학교 국사 7차	2002	교육인적자원부	H
	ta_m71	중학교 국사 7차	2002	교육인적자원부	M
	H-20(1,2001)1	국사 편찬 위원회	2004	문교부	H
	H-1(1,2006)	국사 편찬위원회 국정도서편찬위원회	2006	교육인적자원부	M
	H-2(2,2006)	국사 편찬 위원회	2006	교육인적자원부	H
	H-5(1,2007)	한국근현대사	2007	금성출판사	H
	H-8(1,2007)	한국근현대사	2007	두산	H
	H-10(4,2007)	한국근현대사	2008	천재교육	H
	H-39(1,2002)	국사 편찬 위원회	2009	교육인적자원부	M
	H-57(1,2016)	한국사	2013	지학사	H
	H-54(2,2016)	한국사	2016	금성출판사	H

Table S8: Full corpus inventory by period.

Note: Titles and publishers are reproduced in the original Korean as recorded in the source archives; romanized representative titles appear in main-text Table 1. N = 67 textbooks spanning 1895–2016 (14 Late Chosŏn, 4 Colonial, 8 Postwar, 19 Authoritarian, 22 Democratic). Level abbreviations: E = Elementary, M = Middle school, H = High school, Pub. = public-school audience, Gen. = general audience in the original metadata. Blank publisher cells indicate that the source publisher is not recorded in the merged corpus.

Appendix C: Text processing pipeline

This appendix documents the OCR, cleaning, and morphological processing steps applied to the corpus before analysis.

C.1 OCR for GEI textbooks

Sixteen textbooks from the Georg Eckert Institute (GEI) were available only as scanned PDFs. We processed these books with DeepSeek-OCR, a 3-billion-parameter vision-language model released by DeepSeek-AI, run in BF16 precision.² The model converts page images into structured markdown while preserving broad document layout, including tables, lists, and heading structure. Pages were processed at 200–300 DPI on GPU infrastructure (CUDA 12.4) with flash attention. Processing time was approximately 10–30 seconds per page.

C.2 Artifact removal

Raw OCR output contained residual HTML tags, repeated punctuation from decorative borders, stray symbols, and LaTeX remnants. A post-processing script removed these artifacts and reduced total character count across the 16 GEI textbooks by 23.3%. The cleaning script is included in the replication archive as `03_preprocessing/clean_ocr_artifacts.py`.

C.3 Morphological analysis

Because Korean is agglutinative, a single lexical concept can appear with many particles and inflectional endings. To reduce that source of noise in token-based analysis, we used `kiwipiepy` (Kiwi) to build a noun-focused text field stored as `processed_text`.

The processing pipeline had seven steps:

1. basic cleaning of URLs, e-mail addresses, and special characters;

²Model card and weights: <https://huggingface.co/deepseek-ai/DeepSeek-OCR>.

2. morphological analysis with Kiwi;
3. part-of-speech filtering to retain only common nouns (NNG) and proper nouns (NNP);
4. compound-noun merging to preserve names and technical expressions;
5. stopword removal using a 679-item Korean stopwords list;
6. length filtering to drop single-syllable tokens unless recognized as proper nouns, with tokens restricted to Hangul and Hanja ranges; and
7. number removal.

Across the full corpus, character count fell from 12,988,638 to 4,746,884, a reduction of 63.5%. The processed corpus contains 67 documents with a mean of 22,406 tokens per document. No documents were dropped.

The unfiltered `full_text` field remains available for analyses that require clause structure or verbal morphology. The `processed_text` field is used for exact term counts and other token-based analyses. The sentence-embedding analyses in the main text use `full_text`, because the embedding model requires natural sentence structure.

Appendix D: Term discovery pipeline

The main text outlines the term discovery pipeline. This appendix gives the full specification.

D.1 Extraction prompt

Each text window was presented to the model with the following Korean-language zero-shot instruction:

이 텍스트에서 ‘우리는 누구인가’, ‘어떤 사람이 한국인인가’, ‘한국이란 무엇인가’와 같은 질문에 답하려는 모든 관련 핵심 용어와 표현을 추출하라.

“From this text, extract all relevant key terms and expressions that answer questions such as ‘Who are we?’, ‘What kind of person is Korean?’, and ‘What is Korea?’”

The prompt then specified seven thematic domains: collective belonging, community consciousness, nation and ethnicity, state and territory, unification/independence/liberation, democracy and political systems, and historical legitimacy and values. The model was instructed to return only a JSON array of concrete nouns and to exclude meta-words such as *yongŏ* (용어, ‘term’), *hangmok* (항목, ‘item’), and *yesi* (예시, ‘example’), as well as any term not present in the source text.

D.2 Adaptive window sampling

For each of the 67 textbooks, we sampled text windows of 4,000 characters. The number of windows per document scaled with document length, with a minimum of 3 and a maximum of 50. Sampling always anchored the beginning, middle, and end of each document before filling the remaining slots with randomly selected positions. This procedure guarantees coverage of opening, core, and closing sections regardless of book length. All sampling used a fixed random seed (`seed = 123`), ensuring that window selection was reproducible across model runs.

Model versions and decoding. Extraction was run through Ollama using the model tags

exaone-deep:32b-q4_K_M (4-bit Q4_K_M quantization), aya-expanse:32b, qwen2.5:32b, and gemma3:27b. The within-EXAONE scale check additionally used exaone3.5:2.4b and exaone3.5:7.8b. All calls employed the same generation settings: `num_ctx` 16,384, `num_predict` 512, and `temperature` 0.3. Because decoding at non-zero temperature is not bit-for-bit deterministic, reproducibility is enforced at the level of the term set rather than the raw generations. The fixed window-sampling seed keeps the inputs constant, while the three-of-four cross-model consensus rule (Section D.5) serves as the stability criterion. A term therefore enters the final set only if it survives independent sampling across model families. The exact model tags and option strings are recorded in the replication package (`04_term_discovery/discovery/`).

D.3 Root word extraction

Raw LLM outputs often contained inflected forms, compounds, or multi-morpheme phrases. KiwiPiePy tokenized each extracted expression and retained the longest noun token tagged as NNG or NNP. For example, ‘민족주의적’ was reduced to ‘민족주의,’ and ‘국가 건설’ to ‘국가.’ This procedure aggregates morphological variants before scoring.

D.4 Scoring formula

After semantic-similarity validation and corpus-frequency filtering, surviving terms were scored by period using the following weighted composite:

$$\text{score} = 0.4 \times \hat{f} + 0.6 \times \hat{s} \tag{1}$$

Here $\hat{f}$ denotes min-max normalized frequency within the period, and $\hat{s}$ denotes the min-max normalized cosine similarity between the term embedding and its source-window embedding, computed using `paraphrase-multilingual-MiniLM-L12-v2`. The heavier weight on similarity prevents frequent but weakly relevant nouns, such as `사람` or `시대`, from dominating

the candidate list. The 40/60 frequency–similarity split and the 4,000-character window are fixed design choices rather than tuned parameters. Because a term enters the final set only through the three-of-four cross-model vote (Section D.5), the consensus list is governed by agreement across model families rather than by the precise weighting. The threshold sensitivity analysis in Section D.6 further shows that the nine-term set remains stable across a range of cut-offs on either dimension.

Scores were then aggregated to the root-word level across periods using weighted frequency (score $\times$ document frequency). This produces a single `total_weighted_frequency` value per root word and model. Terms above an absolute floor of $wf \geq 50$ on this measure were retained as each model’s candidate list. We use an absolute floor rather than a $\mu + 2\sigma$ threshold because the standard-deviation rule is sensitive to high-frequency demonym terms that inflate the distribution in some model runs. The fixed floor therefore yields more comparable cut-offs across models that differ substantially in extraction volume.

D.5 Cross-model consensus

The cross-model rule is intended to control for model-specific biases. EXAONE-Deep is Korean-primary, Aya Expanse is multilingual, Qwen2.5 has substantial Chinese-language training, and Gemma 3 is English-primary. Requiring independent models to identify the same term tests whether a candidate reflects structure in the corpus rather than the vocabulary of any single model.

The four per-model candidate lists were merged. A term entered the final list only if:

- it was selected by at least three of the four models, **and**
- it is a conceptual noun rather than a proper noun. Terms denoting specific historical entities (kingdoms, dynasties, and the country itself) function as referential labels rather than contested conceptual constructs and were excluded regardless of vote count.

This procedure produced the nine identity terms used in all downstream analyses (Table S9).

The rationale for the three-of-four threshold and the proper-noun exclusion is discussed in Appendix A. Three conceptual terms that appeared prominently in earlier EXAONE-only analyses (정치, 공동체, and 사회) were selected by only two of the four models in the cross-family ensemble and were therefore excluded.

Term	Models (of 4)	Avg. weighted freq.	Vote
nation / ethnos (<i>minjok</i> , 민족)	4	470	4/4
state (<i>kukka</i> , 국가)	4	275	4/4
citizen / national (<i>kungmin</i> , 국민)	4	120	4/4
independence (<i>tongnip</i> , 독립)	4	216	4/4
democracy (<i>minjujuŭi</i> , 민주주의)	4	194	4/4
unification (<i>t'ongil</i> , 통일)	4	223	4/4
liberation (<i>haebang</i> , 해방)	3	113	3/4
history (<i>yŏksa</i> , 역사)	3	177	3/4
territory (<i>yŏngt'o</i> , 영토)	3	142	3/4

Table S9: Final nine identity terms with cross-model agreement.

Note: *Models (of 4)* = number of 32B-class model families that independently selected the term at $wf \geq 50$. Avg. weighted frequency is the mean `total_weighted_frequency` across the models that selected the term.

D.6 Threshold sensitivity

Two design choices govern the size of the final set. The first is the per-model weighted-frequency floor ($wf \geq 50$). The second is the cross-model voting rule (at least three of four). Both are defensible in isolation. The question is whether either has a substantial effect at the margin. We re-derived the 3-of-4 conceptual consensus from the four-model ensemble’s `phrase_analysis.csv` outputs (EXAONE-Deep-32B, Aya Expanse-32B, Qwen2.5-32B, and Gemma 3-27B) across a range of wf floors, then held wf fixed at 50 and varied the voting rule. Both checks are descriptive. They ask which terms the pipeline would surface under alternative thresholds, holding all other steps (sampling, scoring, similarity validation, proper noun exclusion) constant.

Table S10 reports the wf -floor sensitivity. Lowering the floor expands the consensus set rapidly. At $wf \geq 25$ the formal 3-of-4 rule admits 23 conceptual terms. At $wf \geq 10$ it admits

69. The terms entering at lower floors are general historical, social, and political vocabulary, not identity-construction concepts. Examples include 사회 (society), 공동체 (community), 문화 (culture), 경제 (economy), 정치 (politics), 농민 (peasants and farmers), 양반 (*yangban* aristocracy), 왕권 (royal authority), 토지 (land), 항일 (anti-Japanese), and 의병 (*uibyŏng* righteous-army militias). Raising the floor shrinks the set. At $wf \geq 75$ the consensus drops 역사, 영토, and 해방, all of which appear in the final list at 3/4. The published $wf \geq 50$ floor is the highest at which all nine final terms remain and the lowest at which the only additional 3-of-4 entrant is a general-register word (중앙) rather than an identity-construction concept.

wf floor	Conceptual terms at 3/4 vote	Of 9 final list retained	Notable additions or losses
≥ 10	69	9/9	adds 자유, 평화, 혁명, 전통, 민중, 문화, 경제, 토지, ...
≥ 25	23	9/9	adds 사회, 공동체, 정치, 정체성, 농민, 양반, 왕권, 의병, ...
≥ 50	10	9/9	adds 중앙
≥ 75	6	6/9	loses 역사, 영토, 해방
≥ 100	5	5/9	loses 국민, 역사, 영토, 해방

Table S10: Weighted-frequency sensitivity at fixed 3-of-4 voting (proper nouns excluded). *Note:* The published $wf \geq 50$ row contains all nine final terms. One additional conceptual term, 중앙 (“central”), reaches 3/4 at this floor but was excluded from the final list as general administrative vocabulary rather than an identity-construction concept. Lower floors admit progressively more general vocabulary; higher floors drop terms in the final list.

The vote-rule sensitivity is parallel. Holding $wf \geq 50$ and relaxing the rule from 3-of-4 to 2-of-4 raises the conceptual set from 10 to 14. The four new entrants are 공동체 (community), 불교 (Buddhism), 정체성 (identity/stagnation), and 정치 (politics), a mix of general historical and political vocabulary. Tightening from 3-of-4 to 4-of-4 (unanimous) drops three terms from the final list. 역사 and 해방 fall below threshold for Gemma 3, and 영토 falls below threshold for Qwen2.5. Tightening also drops the non-identity 3/4 term 중앙, leaving six conceptual terms. Unanimous voting is therefore too restrictive. It excludes three recurring identity concepts (history, liberation, and territory), and the omissions would

be substantively costly. The published nine are stable across a narrow band on either dial.

Note on 정체성 (identity). An early version of the analysis included 정체성 through an always-include rule based on its theoretical relevance. Subsequent inspection showed that in this corpus the term overwhelmingly refers to 停滯性 (stagnation), a historiographical concept used to counter Japanese colonial claims about Korean historical backwardness, rather than 正體性 (identity). It was therefore excluded from the final term set and from main-text results. Appendix F retains it as an illustrative case in the full-timeline robustness check, where the homophone artifact produces a misleading positive delta that the postwar-only specification reverses.

Appendix E: Term frequencies by education level

Table S11 disaggregates the period-level frequency analysis by school level: Elementary, Middle School, High School, and Pre-modern for texts that predate the modern school system. Frequencies are reported per 1,000 tokens in `processed_text` and averaged across textbooks within each period-level cell. Standard deviations appear in parentheses. Cells with 0.00 indicate that the term was absent from all textbooks in that cell.

Term	Level	Period				
		Late Chosŏn	Colonial	Postwar	Authoritarian	Democratic
nation (<i>minjok</i> , 민족)	Elem.	0.00	—	8.25 (6.21)	12.40 (11.09)	13.22 (4.84)
	Middle	—	—	6.50 (0.33)	11.74 (7.12)	10.89 (3.29)
	High	—	—	5.65 (3.37)	6.79 (4.14)	7.62 (3.93)
	Pre-mod.	0.05 (0.17)	0.00	4.75 (3.16)	—	—
state (<i>kukka</i> , 국가)	Elem.	0.84 (0.53)	—	6.95 (3.28)	4.88 (4.18)	2.23 (0.11)
	Middle	—	—	7.19 (1.60)	7.21 (2.02)	4.80 (1.48)
	High	—	—	6.14 (1.67)	5.95 (1.89)	4.31 (1.51)
	Pre-mod.	0.47 (0.56)	1.17 (0.58)	3.15 (0.89)	—	—
citizens (<i>kungmin</i> , 국민)	Elem.	0.00	—	14.78 (1.03)	7.36 (7.48)	3.06 (1.12)
	Middle	—	—	3.20 (0.04)	2.87 (1.75)	2.26 (1.37)
	High	—	—	1.18 (1.48)	2.21 (1.25)	2.29 (1.36)
	Pre-mod.	0.05 (0.09)	0.17 (0.20)	1.85 (1.73)	—	—
democracy (<i>minjujuui</i> , 민주주의)	Elem.	0.00	—	9.10 (2.39)	2.33 (4.22)	0.39 (0.23)
	Middle	—	—	0.67 (0.62)	0.13 (0.12)	0.45 (0.37)
	High	—	—	0.09 (0.03)	0.38 (0.38)	0.76 (0.51)
	Pre-mod.	0.00	0.00	0.03 (0.04)	—	—
independence (<i>tongnip</i> , 독립)	Elem.	0.38 (0.66)	—	6.21 (0.27)	4.78 (4.02)	11.34 (6.70)
	Middle	—	—	4.76 (0.12)	6.29 (6.79)	5.47 (4.33)
	High	—	—	2.55 (2.84)	3.94 (4.33)	5.39 (3.64)
	Pre-mod.	0.14 (0.30)	0.34 (0.39)	1.40 (0.47)	—	—
unification (<i>t'ongil</i> , 통일)	Elem.	0.45 (0.61)	—	5.36 (1.15)	3.78 (4.03)	3.39 (1.27)
	Middle	—	—	4.36 (0.67)	4.91 (2.42)	4.38 (1.24)
	High	—	—	2.98 (0.47)	2.98 (1.27)	3.03 (0.66)
	Pre-mod.	0.36 (0.40)	2.63 (3.04)	2.04 (1.10)	—	—
history (<i>yŏksa</i> , 역사)	Elem.	1.10 (0.16)	—	3.92 (1.01)	5.18 (3.41)	4.04 (2.13)
	Middle	—	—	3.79 (1.36)	3.91 (0.63)	4.71 (2.63)
	High	—	—	1.88 (0.26)	2.13 (0.33)	3.08 (1.00)
	Pre-mod.	0.47 (0.43)	0.62 (0.58)	1.62 (0.03)	—	—
territory (<i>yŏngt'o</i> , 영토)	Elem.	0.49 (0.63)	—	0.81 (0.05)	0.75 (0.77)	1.27 (1.00)
	Middle	—	—	1.60 (0.24)	1.08 (0.46)	1.27 (0.46)
	High	—	—	0.39 (0.02)	0.85 (0.59)	0.58 (0.36)
	Pre-mod.	0.30 (0.59)	2.83 (2.70)	0.99 (0.42)	—	—
liberation (<i>haebang</i> , 해방)	Elem.	0.05 (0.08)	—	1.24 (0.87)	0.35 (0.66)	0.00
	Middle	—	—	0.84 (0.05)	0.20 (0.12)	0.36 (0.05)
	High	—	—	0.45 (0.35)	0.27 (0.26)	0.35 (0.20)
	Pre-mod.	0.10 (0.20)	0.00	0.57 (0.28)	—	—

Table S11: Mean frequency per 1,000 tokens by term, period, and education level.

Note: Standard deviations are in parentheses. “—” indicates that no textbooks at that level are present in the period. Elem. = Elementary; Pre-mod. = Pre-modern texts predating the modern school system.

The level breakdown clarifies several patterns that are muted in the period-level aggregates. *Citizen* is far higher in postwar elementary textbooks (14.78 per 1,000 tokens), more than four times its frequency in postwar high school texts (1.18). Early republican civic socialization concentrated in primary education. *Democracy* follows a similar pattern, 9.10 in postwar elementary texts against only 0.39 in democratic-era elementary texts. Democracy shifts from an explicit pedagogical aspiration to a more taken-for-granted background condition. *Nation* is consistently more frequent in elementary and middle school textbooks than in high school texts across all post-liberation periods. Ethnic-national framing was strongest in materials for younger students. *Territory* peaks in colonial-era pre-modern texts at 2.83, the highest value for any term before liberation, consistent with the territorial emphasis of imperial-era historiography.

Appendix F: Full-timeline vs. postwar-only comparison

The main text focuses on post-liberation transitions (Postwar to Authoritarian to Democratic). This appendix reports the full five-period analysis that motivated that restriction and compares the full-timeline and postwar-only logistic deltas.

F.1 Why restrict to post-liberation periods?

The full-timeline analysis included the Late Chosŏn and Colonial periods alongside the three post-liberation periods. The Colonial-to-Postwar transition showed a mean absolute logistic shift of 1.64, roughly four times larger than either later transition (0.43 and 0.40), with the same pattern on the cPCA axis (2.86 against 1.37 and 0.92). That scale difference reflects a qualitatively different kind of change. The Colonial-to-Postwar transition replaced Japanese imperial historiography with post-liberation nationalist education, a wholesale substitution rather than a shift within a continuous discourse tradition.

Including the early periods also changes the scaling baseline. The logistic axis and the StandardScaler normalization are estimated over all loaded data. Nineteenth-century and colonial texts shift that baseline and can inflate apparent authoritarian-to-democratic differences for terms that are absent or marginal before liberation, such as *democracy* and *liberation*. The postwar-only analysis therefore isolates regime-type differences within the continuous post-liberation discourse more cleanly.

F.2 Logistic delta comparison

Table S12 compares the authoritarian-to-democratic logistic deltas from the full five-period run and the postwar-only run. Both runs use the five-period standardization of the visualization pipeline. Under the per-transition standardization behind Table 1 the post-liberation shifts are identical whether or not the early periods are loaded, so this comparison isolates the contribution of the early-period baseline. Terms are sorted by the magnitude of the

difference.

Term	Δ_{full}	Δ_{pw}	Diff.	Label
state (<i>kukka</i> , 국가)	+0.318	+0.289	-0.029	Robust
citizens (<i>kungmin</i> , 국민)	+0.442	+0.390	-0.052	Robust
independence (<i>tongnip</i> , 독립)	+0.376	+0.324	-0.052	Robust
nation (<i>minjok</i> , 민족)	+0.397	+0.295	-0.102	Robust
history (<i>yŏksa</i> , 역사)	+0.440	+0.185	-0.255	Robust
territory (<i>yŏngt'o</i> , 영토)	+0.317	+0.042	-0.275	Robust
unification (<i>t'ongil</i> , 통일)	+0.445	+0.099	-0.346	Robust
liberation (<i>haebang</i> , 해방)	+0.420	-0.027	-0.447	Flat
democracy (<i>minjujuŭi</i> , 민주주의)	+0.485	-0.040	-0.525	Reversed-to-zero

Table S12: Authoritarian-to-democratic logistic deltas for the nine identity terms: full timeline vs. postwar only.

Note: Δ_{full} = delta from the five-period analysis. Δ_{pw} = delta from the postwar-only analysis. Difference = $\Delta_{\text{pw}} - \Delta_{\text{full}}$.

The comparison sorts the terms into three groups.

Robust (7 terms). The authoritarian-to-democratic shift remains positive in both analyses. These terms show a regime-linked difference that is not an artifact of the baseline. Their semantic contexts change when political institutions change, regardless of whether the early periods are included. The magnitude is smaller in the postwar-only run, which is what we would expect once the pre-liberation baseline is removed, but the direction is stable.

Flat (1 term). *Liberation* shows a positive delta in the full run but a near-zero delta in the postwar-only run. Its full-run signal is attributable mainly to the long historical baseline rather than to authoritarian-to-democratic change. In the post-liberation sequence, liberation discourse remains essentially constant.

Reversed-to-zero (1 term). *Democracy* shows a positive delta in the full run but essentially zero in the postwar-only run, indicating that authoritarian and democratic textbooks use democracy in closely overlapping semantic contexts. The authoritarian regime did not remove democratic vocabulary from the textbooks. It appropriated it. The logistic classifier therefore has little leverage for distinguishing the two periods on this term once the early

baseline is removed.

A separate diagnostic case, the homophone 정체성 (which in this corpus refers overwhelmingly to 停滯性 *stagnation* rather than 正體性 *identity*), was excluded from the main analysis on theoretical grounds and is documented in Appendix A’s rejected-terms panel.

F.3 Implications

The comparison shows why period-restricted robustness checks matter in long-horizon semantic analysis. When the baseline includes qualitatively different discourse traditions, the resulting deltas can mix foundational rupture with the within-tradition adjustments we actually want to study. All findings in the main text therefore use the postwar-only analysis. The full-timeline results remain useful as historical context, especially for demonstrating that the largest rupture in the corpus comes at liberation rather than at later regime transitions.

Appendix G: Exemplar sentences

For each identity term discussed in the main analysis, we selected sentences from the extreme poles of all three semantic axes: contrastive variance (cPCA), discriminative (logistic regression), and alignment residual (Procrustes). These are the sentences whose embeddings (KoSimCSE-roberta, 768 dimensions) project most strongly in each direction along each axis. Examples already quoted in the main text are not repeated here.

The three axes capture different geometric properties of semantic change and sometimes place the same sentence at opposite poles. We note these reversals where they occur. Source textbook years are given in parentheses. The full exemplar set with axis coordinates and book identifiers is in the replication archive.

G.1 국민 (*kungmin*, citizen)

Postwar to Authoritarian. *Contrastive variance.* At the Postwar-direction pole, *kungmin* appears in the language of traditional governance:

국왕의 정책 결정이 잘못되었을 때에는 사간원이 간언을 올릴 수 있었고, 국민들에게도 정치적 의견을 제시할 수 있도록 상소의 길이 열려 있었다.

“When royal policy decisions were mistaken, the Saganwŏn could submit remonstrance, and a path for petitions was open so that citizens could present political opinions as well.” (1981)

At the Authoritarian-direction pole, the National Education Charter dominates. All three axes place the Charter at this pole.

국민 교육 헌장

“National Education Charter.” (1981)

Discriminative. The logistic axis separates factional politics from civic improvement. At the Postwar pole:

당 이외에는 아무 것도 보이지 않아 임진왜란 중에 쫓겨 가면서도 이 싸움이요, 남한산성에 포위를 당한 중에도 이 분열이니 그들의 눈에는 국가의 장래와 국민의 생활을 돌아볼 여가가 없었다.

“They could see nothing but their faction. Even while fleeing during the Imjin War, it was still this fighting. Even while besieged at Namhansansŏng, it was still this division. They had no time to consider the nation’s future or the citizens’ livelihood.”

At the Authoritarian pole:

한편, 일부 학자들 사이에는 학문 연구가 국민 생활을 향상시키는 데 기여해야 한다는 생각을 가지고 새로운 학문적 분위기를 만들어 나가기도 하였다.

“Meanwhile, some scholars held the view that academic research should contribute to improving citizens’ lives, and worked to create a new scholarly atmosphere.” (1981)

Alignment residual. Procrustes agrees with cPCA: the Charter at one pole, and at the other, a democratic-era framing of citizens’ rights:

오늘날에는 국민의 자유와 권리가 이전보다 더 잘 보장되고, 국민의 여러 가지 주장이 자유롭게 발표되고 있으며, 각 정당들의 활동도 활발해졌다.

“Today, citizens’ freedoms and rights are better guaranteed than before, citizens’ various opinions are freely expressed, and political party activities have become vigorous.” (1987)

Authoritarian to Democratic. *Contrastive variance.* The negative pole (Authoritarian direction) is dominated by democratic-era sentences about economic hardship:

외환 위기로 인한 국민들의 고통은 엄청난 것이었다.

“The suffering of the citizens caused by the foreign exchange crisis was enormous.” (2008)

The positive pole sentences are textbook index tables rather than narrative prose, which is itself diagnostic. *Kungmin* in democratic textbooks appears so frequently in organizational apparatus that its highest-scoring pole projections are structural rather than discursive.

Discriminative. The logistic axis captures a different contrast. At the Authoritarian pole, *kungmin* appears in European diplomatic history:

그러나, 다시 오스트리아가 알바니아를 세움으로써 뜻을 이루지 못하게 되자, 오스트리아에 대한 국민 감정이 극도로 악화되었다.

“However, when Austria established Albania and thus thwarted [Serbia’s] ambitions, national sentiment toward Austria deteriorated to the extreme.” (1973)

At the Democratic pole, it enters environmental-policy language:

소득 수준뿐만 아니라, 국민 개개인의 삶의 질이 세계화하고 일류화하는 환경 모범 국가를 목표로 하고 있다.

“The goal is an environmentally exemplary state in which both income levels and the quality of life of individual citizens become globalized and world-class.” (2002)

Alignment residual. Procrustes reverses cPCA. The 외환 위기 sentence appears at the positive pole, and the index tables at the negative. The two axes identify the same contrast but orient it differently in the embedding space.

G.2 독립 (*tongnip*, independence)

Postwar to Authoritarian. *Contrastive variance.* The Postwar-direction pole frames independence as armed struggle:

이 밖에도 우리의 독립군이 여러 곳에서 일본군을 괴롭혔다.

“Besides this, our independence army harassed the Japanese military in many places.” (1981)

The top-ranked Authoritarian-direction sentence is quoted in the body of the paper. The remaining pole is dominated by short institutional labels for the Independence Club and the *Tongnip Sinmun*:

최초의 본격적인 민간지는 독립 신문이었다.

“The first fully-fledged private newspaper was the Independent.” (1987)

The cPCA axis separates armed military resistance at the Postwar pole from civic-institutional reform at the Authoritarian pole.

Discriminative. The logistic axis separates a Postwar chapter heading, *“Conservative and Independence factions,”* from architectural and cultural landmarks associated with the independence movement:

독립문과 덕수궁 안의 석조전, 종현 성당(명동 성당) 등 근대식 건물도 세워졌다.

“Modern buildings such as the Independence Gate, the Sökchojŏn Hall inside Töksugung Palace, and the Chonghyŏn Cathedral (Myŏngdong Cathedral) were also constructed.” (1973)

Alignment residual. Procrustes reverses the cPCA poles. At the positive pole, where cPCA placed institutional reform, Procrustes places armed resistance:

청산리 대첩(민족 기록화) 김좌진이 지휘하는 독립군이 일본군과 혈전을 벌여 대승리를 거두었다.

“Battle of Ch’öngsanri (national historical painting): The independence army commanded by Kim Chwa-jin fought a bloody battle against the Japanese military and achieved a great victory.” (1981)

This reversal means that the directions of maximum variance difference and maximum alignment residual are oriented differently. The cPCA axis separates the range of contexts (armed struggle versus institutional reform), while the Procrustes axis captures a structural relocation that runs in the opposite direction.

Authoritarian to Democratic. *Contrastive variance.* At the negative pole (Authoritarian direction), democratic textbooks generalize independence into a global frame:

힘이 약한 세계 여러 민족은 스스로 힘을 모아 독립을 얻었다.

“The various weak nations of the world gathered their own strength and achieved independence.” (1995)

Discriminative. The logistic axis captures an entirely different sense of *tongnip*: not national independence but economic independence.

수공업자들의 독립 현상은 주로 놋그릇, 농기구, 모자, 장도 제조 분야에서 두드러졌다.

“The independence [i.e., independent operation] of artisans was especially prominent in the manufacture of brass vessels, agricultural tools, hats, and ornamental knives.” (1987)

The logistic classifier treats this non-political sense as maximally characteristic of the Authoritarian period, which suggests that authoritarian textbooks more often used *tongnip* in descriptive economic contexts while democratic textbooks reserved the word for political independence.

G.3 민족 (*minjok*, nation)

Postwar to Authoritarian. *Contrastive variance.* Both top-ranked cPCA pole sentences are quoted in the body of the paper. The second-ranked Authoritarian-direction pole continues the colonial-grievance vocabulary:

그러나, 1940년대에 들어서는 민족성 말살 정책 때문에 이러한 활동마저 중단되어, 문화의 암흑기로 접어들게 되었다.

“However, entering the 1940s, even these activities were halted because of the policy to eradicate national identity, ushering in a dark age of culture.” (1973)

Discriminative. The logistic axis captures an economic dimension absent from cPCA. At the Authoritarian pole:

그리고 각종의 근대적 회사와 공장, 은행 등을 설립하여 민족 자본에 의한 근대적 경제 체제를 이룩하려는 노력을 기울였다.

“And [they] made efforts to establish a modern economic system based on national capital by founding various modern companies, factories, and banks.” (1987)

Alignment residual. The Procrustes negative pole captures the national division narrative:

군사적으로 그어진 38선이 정치적 분할선으로 굳어짐으로써 생겨난 우리 나라의 남북 분단은 우리 민족에게 커다란 시련을 안겨 주었다.

“The North-South division of our country, which arose when the militarily drawn 38th parallel hardened into a political dividing line, imposed a great ordeal on our nation.” (1973)

At the positive pole, Procrustes places the growth of modern national consciousness:

근대 의식이 자라면서 교육열이 높아지고 민족 의식이 성장함에 따라, 국어에 대한 관심이 새로와지고 근대적인 역사 의식이 일어나기 시작하였다.

“As modern consciousness grew, educational fervor rose, and national consciousness developed, interest in the national language was renewed and a modern historical consciousness began to emerge.” (1981)

Authoritarian to Democratic. The top cPCA negative pole sentence is quoted in the body of the paper. cPCA and Procrustes agree on this transition, placing the same sentences at the same poles. The cPCA positive pole (Democratic direction) invokes the colonial-era nationalist historiographic tradition:

민족사의 수난, 일제 식민 통치

“The ordeals of national history, Japanese colonial rule.” (2007)

Discriminative. The logistic axis identifies a distinctive sentence at the Authoritarian pole that the variance axes miss:

그리하여, 1961년 5월 16일, 박정희 장군을 중심으로 한 군인들은, 나라와 민족을 혼란과 위기에서 구하기 위하여 혁명을 일으켰다.

“Thus, on May 16, 1961, soldiers centered on General Park Chung-hee launched a revolution to save the country and the nation from chaos and crisis.” (1973)

This sentence does not appear at the extremes of either variance axis, yet the logistic classifier identifies it as maximally characteristic of the Authoritarian period. The framing of the 1961 coup as a national salvation act, described without qualification in a textbook produced under the subsequent curriculum, is precisely the kind of period-specific discourse that the discriminative axis is designed to capture.

G.4 해방 (*haebang*, liberation)

The contrastive axes for *haebang* do not separate different political framings of national liberation. Instead, they track the word’s polysemy, separating 1945 liberation from colonial rule, premodern slave emancipation, and social movement liberation. This polysemy structure is what accounts for the term’s exceptionally large Procrustes shift during democratization, discussed in the main analysis.

Postwar to Authoritarian. *Contrastive variance.* At the negative pole (Postwar direction), *haebang* means national liberation from colonial rule, paired with the trauma of subsequent division:

1945년 해방 후에도 국토의 남북 분단과 민주주의 대 공산주의의 싸움이라는 세계적인 현실을 무릅쓰지 않을 수 없었다.

“Even after liberation in 1945, [Korea] could not escape the global reality of the North-South division of the homeland and the struggle between democracy and communism.” (1973)

At the positive pole (Authoritarian direction), the word refers to premodern slave emancipation:

그는 먼저 노비 안검법을 실시하여 호족들이 불법으로 소유하고 있는 많은 노비를 해방시켰다.

“He first implemented the Slave Review Act and emancipated the many slaves illegally held by powerful clans.” (1973)

Discriminative. At the Postwar pole, *haebang* names the event directly as a chapter heading: “*Our country’s liberation*” (우리 나라의 해방, 1954). At the Authoritarian pole, the word appears within cultural-history narratives rather than as a standalone political category.

Alignment residual. Procrustes reverses cPCA, placing slave emancipation at the negative pole and national liberation at the positive. This reversal confirms that the two senses of the word occupy distinct positions in the embedding space, and the axes rotate through that space differently.

Authoritarian to Democratic. *Contrastive variance.* The polysemy structure persists but rotates. At the negative pole (Authoritarian direction), the word again means premodern emancipation:

공민왕은 신돈을 등용하여, 권문세족이 빼앗은 토지를 원래의 주인에게 돌려주고 불법적으로 소유한 노비를 해방시켰다.

“King Kongmin appointed Sindon, returned land seized by the powerful families to its original owners, and emancipated slaves held illegally.” (2006)

At the positive pole (Democratic direction), the word enters a social movement vocabulary absent from authoritarian textbooks:

여성 운동 단체들은 여성의 사회적 지위 향상과 여성 해방을 위한 사회 운동을 전개하였다.

“Women’s movement organizations carried out social movements for the improvement of women’s social status and women’s liberation.” (2016)

Discriminative. The logistic axis captures a different contrast. At the Authoritarian pole, *haebang* means 1945 national liberation:

민족은 해방되었으나 고대하던 독립이 즉시 이루어지지 않는 않았다.

“The nation was liberated, but the long-awaited independence was not immediately achieved.” (1973)

At the Democratic pole, it appears within the nationalist-socialist split:

사회주의 사상이 유입되고 그 세력이 급속히 확산되면서 독립 운동계는 민족 해방을 부르짖는 민족주의 계열과 계급 해방을 목표로 하는 사회주의 세력으로 양분되었다.

“As socialist thought flowed in and its influence spread rapidly, the independence movement was divided into a nationalist faction calling for national liberation and a socialist faction aiming for class liberation.” (2007)

Across both transitions, the contrastive axes for *haebang* separate senses of the word rather than political framings of the same event. The modern Korean sense of the term, liberation from colonial rule, appears at similar positions regardless of period. What shifts is the word’s secondary usage: premodern slave emancipation in authoritarian textbooks gives way to social movement liberation (women’s, class) in democratic ones.

G.5 사회 (*sahoe*, society)

Postwar to Authoritarian. *Contrastive variance.* At the Postwar-direction pole, *sahoe* appears as a chapter-heading label:

근대 사회의 성장

“Growth of Modern Society.” (1987)

At the Authoritarian-direction pole, the term appears in a concrete description of premodern social hierarchy:

따라서, 모든 정치는 왕과 귀족이 주도하였고, 신라의 골품 제도와 같은 엄격한 신분 제도에 따라 사회가 유지되었다.

“Accordingly, all politics were led by kings and aristocrats, and society was maintained according to strict status systems such as Silla’s bone-rank system.” (1987)

Discriminative. The logistic axis captures a completely different usage. At the Postwar pole, *sahoe* appears not in Korean social history but in a description of the United Nations:

사무 총장은, 안전 보장 이사회에서 사람을 골라서, 총회에서 반수 이상의 추천을 얻어야 한다.

“The Secretary-General must be selected by the Security Council and receive the recommendation of more than half the General Assembly.” (1954)

At the Authoritarian pole, the term appears in Confucian moral language:

조선 사회에서는 유교적인 가족 질서를 바탕으로 충효의 윤리가 강조되었다.

“In Chosŏn society, the ethics of loyalty and filial piety were emphasized on the basis of Confucian family order.” (1987)

The logistic axis thus identifies the term’s postwar usage in international-institutional contexts as discriminative, a usage that disappears under authoritarianism.

Alignment residual. Procrustes agrees with cPCA at the negative pole (same chapter-heading sentence). At the positive pole, Procrustes places a shorter Confucian-society label:

유교 중심의 양반 사회

“Confucian-centered yangban society.” (1973)

Authoritarian to Democratic. *Contrastive variance.* The negative pole (Authoritarian direction) brings the nationalist-socialist alliance into view:

한편, 단일화된 민족 운동을 위한 노력이 계속 추진되어 민족주의계와 사회주의계가 합작하여 신간회를 조직하였다.

“Meanwhile, efforts toward a unified national movement continued, and the nationalist and socialist factions jointly organized the Singanhoe.” (1981)

At the positive pole (Democratic direction):

옛날보다 지금의 사회 제도는 더 복잡해졌다.

“Compared to the past, today’s social institutions have become more complex.” (1973)

Alignment residual. Procrustes reverses cPCA for this transition. The Singanhoe sentence appears at the Procrustes *positive* pole, and the “more complex” sentence at the Procrustes *negative* pole. The two axes identify the same semantic contrast but orient it in opposite directions, indicating that the dimension of maximum variance difference and the dimension of maximum structural reorganization are not aligned for *sahoe* in this transition.

G.6 통일 (*t'ongil*, unification)

Postwar to Authoritarian. *Contrastive variance.* At the Postwar-direction pole, *t'ongil* refers to pre-modern historical unification:

3층 석탑(석가탑) 통일 신라 시대의 대표적인 3층 석탑이다.

“Three-story stone pagoda (*Sökkat’ap*): A representative three-story stone pagoda of the Unified Silla period.” (1981)

The Authoritarian-direction pole is the National Education Charter:

길이 후손에 물려줄 영광된 통일 조국의 앞날을 내다보며, 신념과 긍지를 지닌 근면한 국민으로서, 민족의 슬기를 모아 즐기찬 노력으로, 새 역사를 창조하자.

“Looking ahead to the future of a glorious unified fatherland to be bequeathed to posterity, let us, as diligent citizens with conviction and pride, gather our nation’s wisdom and create a new history through persistent effort.” (1991)

Discriminative. The logistic axis separates two ways of referencing Unified Silla. At the Postwar pole, it appears within evaluative cultural claims:

삼국 시대와 통일 신라 시대의 찬란한 불교 예술이 그것이며, 고려 시대의 자기와 활자, 조선 시대의 측우기나 한글 같은 것은 어느 나라에 비할 바가 없이 훌륭한 것이다.

“The splendid Buddhist art of the Three Kingdoms and Unified Silla periods, the celadon and movable type of the Koryŏ period, and the rain gauge and Hangŭl of the Chosŏn period are all magnificent beyond comparison with any other country.” (1963)

At the Authoritarian pole, it appears in descriptive captions:

포석정 통일 신라 시대의 건축물로, 돌로 도랑 모양을 만들어 물이 흐르게 하였다.

“*P’osŏkchŏng*: A Unified Silla structure in which channels were carved into stone to let water flow.” (1987)

Alignment residual. Procrustes reverses cPCA. The Charter appears at the *negative* pole, while Buddhist material culture appears at the positive pole. As with other terms, the reversal indicates that variance expansion and structural repositioning operate along different geometric axes.

Authoritarian to Democratic. *Contrastive variance.* The Authoritarian-direction pole confines *t'ongil* to historical reference:

통일 신라의 왕권 강화

“Strengthening of royal authority in Unified Silla.” (1987)

The Democratic-direction pole is contemporary policy:

통일 정책과 남북 대화

“Unification policy and inter-Korean dialogue.” (2007)

The highest-scoring pole exemplar for cPCA is already quoted in the body of the paper.

Discriminative. The logistic axis brings in a global comparative vocabulary absent from both other axes:

특히, 나세르는 수에즈 운하의 국유화를 단행하여 민족의 의지를 보였고, 시리아와 함께 통일 아랍 공화국을 형성하기도 하였다.

“In particular, Nasser demonstrated the will of the nation by nationalizing the Suez Canal, and also formed the United Arab Republic with Syria.” (1973)

This sentence, which uses *t’ongil* in a non-Korean context, is the most discriminative of the Authoritarian period. The logistic classifier treats comparative world-history references to unification as distinctly authoritarian-era language.

Alignment residual. Procrustes again reverses cPCA, placing “Unification policy and inter-Korean dialogue” at the negative pole and “Unified Silla strengthened royal authority” at the positive.

G.7 역사 (*yŏksa*, history)

Postwar to Authoritarian. *Contrastive variance.* The Postwar-direction pole is the National Education Charter, where *yŏksa* is part of the injunction to create new history:

길이 후손에 물려준 영광된 통일 조국의 앞날을 내다보며, 신념과 긍지를 지닌 근면한 국민으로서, 민족의 슬기를 모아 즐기찬 노력으로, 새 역사를 창조하자.

“Looking ahead to the future of a glorious unified fatherland to be bequeathed to posterity, let us, as diligent citizens with conviction and pride, gather our nation’s wisdom and create a new history through persistent effort.” (1990)

At the Authoritarian-direction pole, the term is in a more concrete pedagogical context:

역사를 알아보는 데 이용되는 자료에는 여러 가지가 있다.

“There are many kinds of sources used for investigating history.” (1987)

Discriminative. The logistic axis identifies a different contrast. At the Postwar pole, *yŏksa* appears within a discussion of historiographic persecution:

사필로 말미암은 사람의 화인 까닭에 역사가는 이를 특히 사화라 이른다.

“Because the literati suffered calamity on account of the historian’s brush, historians call these events specifically ‘literati purges.’”

At the Authoritarian pole, history becomes a programmatic declaration:

역사란 지나가 버린 과거의 사실과 연대를 공부하고 기억하는 학문이 아니다.

“History is not a discipline of studying and memorizing the facts and dates of a vanished past.” (1973)

Alignment residual. Procrustes fully agrees with cPCA on this transition, with the Charter at the negative pole and pedagogical sources at the positive.

Authoritarian to Democratic. *Contrastive variance.* At the negative pole (Authoritarian direction), democratic textbooks take a comparative and reflexive approach:

올바른 역사 의식을 가지고 우리 역사를 객관적으로 이해하려고 하면, 우리 역사뿐만 아니라 주변 민족, 주변 사회의 역사도 바르게 이해해야 한다.

“To understand our history objectively with a correct historical consciousness, we must properly understand not only our history but also the history of neighboring nations and societies.” (2004)

Discriminative. The logistic axis captures a different dimension. At the Authoritarian pole, *yŏksa* appears in the narrative of World War II:

이로써 인류 역사상 최대의 전쟁인 제2차 세계 대전이 연합국의 승리로 끝났다.

“Thus the largest war in human history, the Second World War, ended with the Allied victory.” (1973)

Alignment residual. The Procrustes negative pole captures a similar comparative framing to cPCA but with different emphasis:

따라서 역사를 바르게 이해한다는 것은 세계사적 보편성과 지역적 특수성을 균형 있게 파악함을 의미하는 것이다.

“Therefore, understanding history correctly means grasping the balance between world-historical universality and regional particularity.” (2002)

G.8 국가 (*kukka*, state)

Postwar to Authoritarian. *Contrastive variance.* The Postwar-direction pole places *kukka* in a fiscal context:

국가 재정과 농민의 부담

“State finances and the burden on farmers.” (1987)

At the Authoritarian-direction pole, the term is part of narratives of ancient state formation:

삼국은 점차 왕을 중심으로 하는 중앙 집권적인 국가로 발전하였다.

“The Three Kingdoms gradually developed into centralized states centered on the king.”
(1981)

Discriminative. The logistic axis captures a Cold War ideological dimension. At the Postwar pole:

민주주의 국가의 국회와 공산주의 국가의 국회는 어떻게 다른지 비교하여 보자.

“Let us compare how the parliament of a democratic state and the parliament of a communist state differ.” (1963)

This comparative Cold War framing of *kukka* is the most discriminative sentence for the Postwar period, a usage that disappears from authoritarian-era textbooks, where ideological comparison gives way to historical narration.

Alignment residual. Procrustes agrees with cPCA on this transition. The same sentences occupy the same poles.

Authoritarian to Democratic. *Contrastive variance.* At the negative pole (Authoritarian direction), democratic textbooks extend *kukka* into Cold War geopolitics:

제24회 서울 올림픽 대회 개막 행사(1988) 동서 냉전의 갈등으로 한동안 올림픽에 참여하지 않았던 공산권 국가들까지 참가하여 동서 화합의 축제가 되었다.

“Opening ceremony of the 24th Seoul Olympic Games (1988): Even communist-bloc states that had not participated in the Olympics for some time due to East-West Cold War tensions joined, making it a festival of East-West reconciliation.” (2002)

At the positive pole (Democratic direction), *kukka* is a premodern institutional descriptor:

통일 신라는 국가의 수입을 늘리고 귀족 세력을 억제하기 위하여 토지 제도를 개편하였다.

“Unified Silla reformed its land system to increase state revenue and restrain aristocratic power.” (1995)

Discriminative. The logistic axis picks up a different sentence at the Authoritarian pole, a declaration of democratic identity:

우리 나라에는 종교가 여러 가지 있으며 국교가 없이 신앙의 자유가 보장되어 있는 것은, 우리 나라가 민주주의 국가임을 말해 주는 것이다.

“The fact that our country has many religions and guarantees freedom of belief without a state religion shows that our country is a democratic state.” (1973)

Alignment residual. At the Procrustes positive pole, a chronological timeline from 1989 marks the diplomatic opening to Eastern Europe:

1987: 6.29 민주화 선언 1988: 노태우 정부 성립, 제24회 서울 올림픽 대회 1989: 헝가리, 폴란드 등 동구권 국가와 수교

“1987: June 29 Democratization Declaration. 1988: Inauguration of the Roh Tae-woo government, 24th Seoul Olympic Games. 1989: Establishment of diplomatic relations with Hungary, Poland, and other Eastern bloc states.” (2001)

At the Procrustes negative pole, *kukka* appears in the same 15th-century fiscal context as the cPCA positive pole, reversing the orientation.

Additional exemplar sentences for other terms are available in the replication archive.

Appendix H: Per-book contributions and sensitivity

This appendix first records the full specifications of the three semantic axes summarized in the Methods, then reports per-book robustness checks on the pooled-axis results in the Results section of the main text. The per-book checks address two questions: where individual textbooks sit on each cPCA axis (per-book projection onto the axes used to produce main-text Table 2), and whether any single textbook moves the axis directions themselves (leave-one-book-out jackknife). The methodology mirrors the Results section of the main text, with one difference. This pipeline standardizes embeddings over the full five-period sentence set, while the main-text axes standardize per transition, which is why per-book full-data shifts match main-text Table 2 closely (within 0.08) rather than exactly. The axes are otherwise computed identically on pooled within-transition sentences with the same RNG seed (2025), with only the projection step per-book. Scripts and outputs are in the replication package under `05_embeddings/`.

Semantic-axis specifications

For each adjacent-period transition we fit three complementary axes on the pooled, standardized sentence embeddings.

The discriminative (logistic regression) axis measures how cleanly the two periods can be told apart. We pool sentences from both periods across all nine terms and fit a binary L2-regularized logistic regression ($C=1.0$, balanced class weights) to predict period membership. The unit-normalized coefficient vector $\hat{w}$ is the discriminative axis. The per-term shift is the projected distance between the term’s two period centroids, $(\bar{x}_{\text{later}} - \bar{x}_{\text{earlier}}) \cdot \hat{w}$, positive toward the later period. As a separability diagnostic we also fit a per-term classifier and report its 5-fold cross-validated accuracy.

The contrastive-variance (cPCA) axis measures where the later period’s embeddings occupy a wider range of positions than the earlier period’s, following Abid et al. (2018). We

compute the sentence-embedding covariance separately for each period (pooled across all nine terms) and take the top eigenvector of $\Sigma_{\text{later}} - \alpha \Sigma_{\text{earlier}}$, with $\alpha = \text{tr}(\Sigma_{\text{later}})/\text{tr}(\Sigma_{\text{earlier}})$ set from the data so the comparison is symmetric in scale. The axis is sign-pinned so the pooled later-period centroid sits on the positive side. It supports two independent measurements: a per-term centroid shift (how far the term’s average sentence moved) and a per-term SD ratio, $\text{sd}(\text{later})/\text{sd}(\text{earlier})$ of sentence projections (whether the spread of the term’s own sentences grew). A term can move along the axis without widening, as in the *liberation* case. Because cPCA picks variance-growth directions by construction, SD ratios are biased above 1. To detect true compression we also compute direction-free measures (the trace ratio of full period covariances, the top eigenvalue of a reverse-cPCA fit, and the participation ratio $\text{PR}(\Sigma) = (\text{tr } \Sigma)^2/\text{tr}(\Sigma^2)$), detailed in Section H.6.

The alignment-residual (Procrustes) axis measures the structural relocation that survives after the two periods’ embeddings are rotated into closest alignment. Adapting Hamilton et al. (2016), we center each period’s embeddings, subsample the larger set to match the smaller, and obtain the optimal rotation via SciPy’s `orthogonal_procrustes`. The axis is the top right-singular vector of the residual matrix between the aligned embeddings. Under leave-one-book-out resampling it is sample-noise-dominated (median angular displacement $\approx 86^\circ$ versus 6° for the other two axes; Section H.4), so we do not interpret per-term Procrustes shifts and retain the axis only as a third structural diagnostic.

H.1 Drop-one-book sensitivity

For each (term, transition, axis), we recomputed the sentence-weighted shift after removing each textbook’s sentences in turn. Table S13 reports the cPCA shifts whose magnitude moved by 0.30 or more under any single removal, sorted by the magnitude of the largest single-book delta.

Term	Transition	Full	After drop	Δ	Worst book	Sign flip?
Democracy (민주주의)	Auth–Dem	+0.73	−0.39	−1.12	ta_e52	Yes
Democracy (민주주의)	Post–Auth	+0.36	+1.31	+0.95	ta_e52	No
Independence (독립)	Post–Auth	+3.22	+2.44	−0.78	ta_m21	No
Liberation (해방)	Post–Auth	−1.68	−2.30	−0.62	ta_h31	No
Nation (민족)	Post–Auth	+2.03	+2.55	+0.52	ta_h21	No
Liberation (해방)	Auth–Dem	−0.16	−0.60	−0.44	H-5(1,2007)	No
Territory (영토)	Auth–Dem	−0.71	−0.33	+0.38	ta_h32	No
Citizens (국민)	Post–Auth	+2.15	+2.51	+0.37	ta_e52	No
History (역사)	Auth–Dem	−1.46	−1.82	−0.37	ta_e51	No
Unification (통일)	Auth–Dem	+2.53	+2.17	−0.36	H-6(1,2007)	No

Table S13: Drop-one-book sensitivity for cPCA shifts.

Note: *Full shift* is the per-book pipeline’s full-data value, which matches main-text Table 2 to within 0.08 (the pipelines differ in standardization scope). *After drop* is the shift after removing the most-influential single book. Δ is the difference. Only one shift in the analysis sign-flips on single-book removal: *democracy* Authoritarian-to-Democratic, where ta_e52 (1987 elementary 5th-curriculum *sahoe*) carries enough of the centroid that its removal reverses the shift direction.

The remaining eight cPCA shifts in main-text Table 2 change by less than 0.30 in magnitude under any single-book removal. The most fragile result is the *democracy* Auth–Dem shift, which sign-flips when ta_e52 is removed. We note this in the main text and treat the democratic-era *democracy* expansion as conditional on a single elementary school text that carries the centroid for that term. The other large Δ values reflect concentration of substantive content in particular books rather than fragility of the underlying claim: ta_m21 (1963 middle school 2nd-curriculum) anchors the Postwar baseline for *independence*, and ta_h31 (1973 Yushin high school) anchors authoritarian-era *liberation*.

H.2 Within-period sub-era heterogeneity

We split the Authoritarian period into Yushin (1973–1979), Chun (1980–1987), and Roh transition (1988–1993) sub-eras, and the Democratic period into 6th-curriculum NIKH (1995–2001) and 7th-and-later *kŏmjŏngje* (2002–) sub-eras. For each sub-era and each term, we compute the sentence-weighted mean projection on the corresponding cPCA axis. The val-

ues reported are absolute positions on the axis, not shifts. Larger positive values within a sub-era indicate textbooks positioned further in the direction the cPCA axis was sign-pinned to make positive (the dominant pooled-shift direction within each transition).

Term	Yushin (1973–79)	Chun (1980–87)	Roh transition (1988–93)
Independence (독립)	+4.02	+5.74	+1.18
Nation (민족)	+2.37	+3.93	+1.33
Citizens (국민)	−0.41	+0.32	−1.11
History (역사)	+0.07	+0.73	+0.03
Democracy (민주주의)	−2.06	−2.35	—
Liberation (해방)	−0.87	−3.85	−3.64
State (국가)	−5.20	−5.51	−7.51
Territory (영토)	−6.37	−5.91	−7.50
Unification (통일)	−5.51	−5.33	−6.26

Table S14: Authoritarian sub-era cPCA centroids on the Postwar-to-Authoritarian axis.

Note: Larger positive values indicate stronger movement in the dominant authoritarian-era direction; *democracy* for Roh is missing because no Roh-transition textbook has sufficient *democracy* sentences for a stable estimate.

The Chun sub-era shows the largest expansion in Independence, Nation, and Citizens: the 1981 4th-curriculum and 1987 5th-curriculum middle school texts carry the regime’s ethnonational vocabulary further than the 1973 Yushin texts that introduced the *kukchǒngje* system. The Roh-transition sub-era softens on Independence, Nation, and Citizens, suggesting a tail-of-authoritarianism softening already underway in 1988–1993.

Term	6th curriculum (1995–2001)	7th+ (2002–)
Unification (통일)	+2.76	+4.97
Citizens (국민)	+1.92	+2.37
Independence (독립)	+1.25	+1.47
Democracy (민주주의)	+4.10	+3.99
Nation (민족)	−1.18	−0.23
State (국가)	+0.87	+0.97
Liberation (해방)	−0.52	+1.11
Territory (영토)	−1.30	−1.97
History (역사)	−8.12	−7.40

Table S15: Democratic sub-era cPCA centroids on the Authoritarian-to-Democratic axis. *Note:* The 6th-curriculum books (1995–2001) were produced under partial state authority before the 2002 institutional shift to *kŏmjŏngje* (certification rather than state authorship). The 7th-and-later books are the first generation of the new privately-published era.

The shift at the 2002 boundary concentrates on three terms: Unification (6th +2.76 vs. 7th +4.97, a 2.2-standard-deviation shift), Liberation (sign-flips between sub-eras), and Territory (further negative under *kŏmjŏngje*, reflecting the introduction of the Northeast-Asia / Dokdo discourse). For the other six terms, the 6th-curriculum and 7th-curriculum sub-eras occupy similar positions on the cPCA axis, indicating that the democratic-era reorganization was complete by 1995 for most of the vocabulary and concentrated at the 2002 institutional boundary only for terms most directly tied to inter-Korean policy and territorial dispute.

H.3 Per-book strip plot

Figure S1 shows the per-book cPCA scores for all nine identity terms across both transitions. Each point is one textbook positioned by its mean cPCA projection (horizontal) and dodged vertically within each (term, transition) cell to prevent overplotting. Dashed lines mark the sentence-weighted period mean reproduced from main-text Table 2.

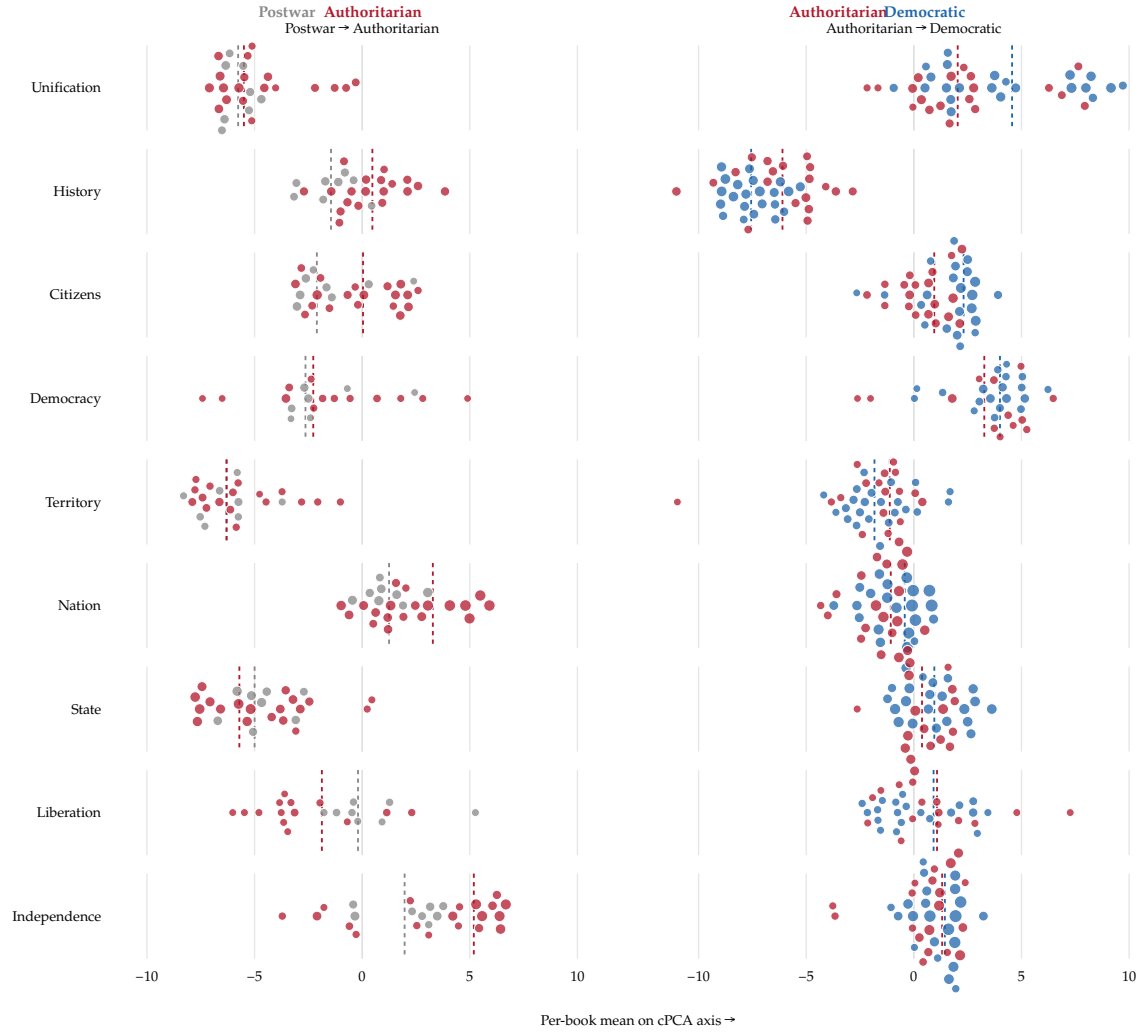


Figure S1: Per-book cPCA scores for the nine identity terms across the two post-liberation transitions.

Note: Each point is one textbook. Dashed lines mark the sentence-weighted period mean reproduced from main-text Table 2. Within-period spread is visible at a glance; particularly tight clusters indicate that the period-level shift is robust to book-level variation, while wider spreads (Liberation, Democracy) flag terms whose period-level estimates are more concentrated in particular textbooks.

H.4 LOBO axis stability

For each transition and axis, we recomputed the axis vector after dropping each book's sentences entirely, and measured the angular displacement (in degrees) between the leave-one-book-out axis and the full-corpus axis. Table S16 reports the median and 90th-percentile

displacements, plus the most influential single book.

Transition	Axis	Median	90th pct.	Max	Most influential book
Post-Auth	LR	6.34	15.28	27.84	ta_h21
Post-Auth	cPCA	4.62	10.24	17.29	ta_m21
Post-Auth	PA	85.94	88.05	89.08	ta_m42
Auth-Dem	LR	4.49	7.89	11.19	ta_h32
Auth-Dem	cPCA	5.58	13.05	18.82	H-6(1,2007)
Auth-Dem	PA	86.75	89.29	89.97	ta_m41

Table S16: Leave-one-book-out angular displacement of axis directions, in degrees.

Note: The Procrustes axis is sample-noise-dominated under perturbation; the median LOBO displacement is approximately 86 degrees, indicating that the axis is rotated to near-orthogonality by random subsampling rather than by any single book’s contribution. The LR and cPCA axes are stable: median displacements 4–6 degrees, with no single book moving the axis by more than 28 degrees.

The Procrustes instability is intrinsic to the alignment procedure (which subsamples one period to match the size of the other) rather than a sign of any single book’s outsize influence. Neither the signs nor the per-term magnitudes of the PA shifts in main-text Table 2 are stable across re-randomizations; we therefore report the PA column for completeness and do not read individual PA values as directional or magnitude claims. The LR and cPCA axes are stable: even H-6(1,2007), the largest single textbook in the corpus and the most influential single book on the Auth-Dem cPCA axis, displaces the axis by only 19 degrees, and no shift in main-text Table 2 sign-flips on its removal.

H.5 Per-period context width on the cPCA axis

The cPCA shift in main-text Table 2 measures centroid movement along the variance-distinctive direction. A complementary direct measure of context width is the standard deviation of sentence projections on the cPCA axis within each period. A ratio greater than 1 indicates that the later period uses the term in a wider distribution of contexts along the cPCA axis than the earlier period.

Term	Postwar to Authoritarian			Authoritarian to Democratic		
	σ_{Postwar}	σ_{Auth}	ratio	σ_{Auth}	σ_{Dem}	ratio
Independence (독립)	5.31	5.25	0.99	3.93	4.66	1.18
Nation (민족)	4.88	5.43	1.11	4.55	5.72	1.26
Citizens (국민)	4.45	5.79	1.30	3.84	4.59	1.19
History (역사)	3.89	4.74	1.22	4.91	4.97	1.01
Democracy (민주주의)	2.68	3.94	1.47	3.13	4.01	1.28
Unification (통일)	3.59	4.48	1.25	4.58	5.65	1.23
State (국가)	3.94	4.76	1.21	3.51	4.60	1.31
Territory (영토)	2.81	3.69	1.31	3.18	3.74	1.18
Liberation (해방)	4.71	4.68	0.99	3.52	4.17	1.18

Table S17: Standard deviation of sentence projections on each cPCA axis, by period.

Note: The ratio column is $\sigma_{\text{later}}/\sigma_{\text{earlier}}$; values greater than 1 indicate widening of the context distribution. All nine terms show context-width widening under democratization (right-most ratios all exceed 1.0); under authoritarianism, seven of nine widen and two (*independence*, *liberation*) hold roughly constant width (ratios 0.99) despite substantial centroid movement.

Read together with main-text Table 2, this table makes one substantive claim of the Findings section direct. Under democratization, every identity term in the analysis is used in a wider distribution of contexts along the cPCA axis. The signed cPCA shifts in main-text Table 2 carry information about *which direction* centroids move along the axis. The SD ratios here carry information about *whether the distribution widens or narrows on that specific axis*. The two measures coincide for most terms, but they diverge for *independence* and *liberation* under authoritarianism (substantial centroid movement with no widening on the cPCA axis) and for *history* and *territory* under democratization (centroids move opposite to the dominant democratic direction while context distributions still widen on the cPCA axis).

The cPCA axis is selected by maximizing target-period variance against background-period variance, so its SD ratio is bounded below by 1 whenever any contrastive direction exists at all. Section H.6 reports three bidirectional measures (total variance, reverse-cPCA top eigenvalue, effective dimensionality) that can detect compression, and shows that the cPCA-axis non-widening of *liberation* is a direction-specific finding rather than a wholesale shrinkage of liberation discourse.

H.6 Bidirectional context-spread measures

The SD ratio in Section H.5 is computed on the cPCA axis, which is constructed to maximize the later period’s variance against the earlier period’s variance. By construction, that ratio is at least 1 whenever any direction satisfies the contrast. It cannot detect compression, only the absence of expansion. To answer the bidirectional question (did any term’s semantic spread shrink between periods, in any sense) we compute three model-free measures from the raw sentence embeddings. All three are sign-convention-free.

1. **Total-variance ratio** $\text{tr}(\Sigma_{\text{later}})/\text{tr}(\Sigma_{\text{earlier}})$. The trace of the embedding covariance is the total spread of the term’s sentence cloud across all dimensions. Values below 1 indicate global compression.
2. **Reverse-cPCA top eigenvalue** $\lambda_{\max}(\Sigma_{\text{earlier}} - \Sigma_{\text{later}})$. Positive values mean some direction exists along which the earlier period had more variance than the later, a compression direction. Reported alongside $\lambda_{\max}(\Sigma_{\text{later}} - \Sigma_{\text{earlier}})$ for symmetry. Whichever is larger identifies the dominant direction of asymmetric variance change.
3. **Effective dimensionality (participation ratio)** $\text{PR}(\Sigma) = \text{tr}(\Sigma)^2/\text{tr}(\Sigma^2)$. The number of dimensions a Gaussian with this covariance effectively varies along, bounded in $[1, d]$. Ratios below 1 indicate that the term’s contexts collapsed onto fewer dimensions of variation.

Term	Transition	N_e	N_l	Variance asymmetry			PR ratio
				trace ratio	λ_{exp}	λ_{comp}	
Independence (독립)	Post-Auth	345	1374	0.953	3.26	3.37	1.03
History (역사)	Post-Auth	276	868	0.982	4.20	2.88	0.87
Citizens (국민)	Post-Auth	330	905	0.983	3.66	3.51	1.04
Nation (민족)	Post-Auth	600	2457	0.989	2.12	2.68	1.01
State (국가)	Post-Auth	560	1687	0.994	2.83	3.65	1.08
Democracy (민주주의)	Post-Auth	127	168	1.022	4.72	5.17	0.99
Liberation (해방)	Post-Auth	73	82	1.046	8.25	6.25	1.01
Unification (통일)	Post-Auth	345	962	1.046	2.55	2.31	1.02
Territory (영토)	Post-Auth	100	255	1.051	4.67	2.94	0.94
Democracy (민주주의)	Auth-Dem	168	516	0.983	4.27	4.74	1.25
Liberation (해방)	Auth-Dem	82	265	0.986	5.23	7.02	1.33
Citizens (국민)	Auth-Dem	905	1847	0.993	2.58	3.81	1.17
Nation (민족)	Auth-Dem	2457	5764	1.021	1.83	1.79	1.09
History (역사)	Auth-Dem	868	2353	1.029	2.21	4.64	1.38
State (국가)	Auth-Dem	1687	2954	1.034	2.41	2.29	1.11
Unification (통일)	Auth-Dem	962	2250	1.036	4.34	2.73	0.97
Independence (독립)	Auth-Dem	1374	4216	1.054	1.99	1.46	1.17
Territory (영토)	Auth-Dem	255	501	1.069	3.92	2.61	1.03

Table S18: Bidirectional context-spread measures computed from raw sentence embeddings, one row per (term, transition).

Note: N_e and N_l are sentence counts in the earlier and later periods. Total-variance ratio below 1 indicates global compression; PR ratio below 1 indicates collapse onto fewer effective dimensions; $\lambda_{\text{compress}} > \lambda_{\text{expand}}$ indicates that the strongest direction of asymmetric variance change is a compression direction.

Three results follow.

First, no term shows large-scale compression in total variance. The trace ratios across all 18 (term $\times$ transition) cells fall in [0.95, 1.07]. Authoritarian-era ratios below 1 (Independence 0.953, History 0.982, Citizens 0.983, Nation 0.989, State 0.994) indicate mild consolidation across most well-sampled terms rather than dramatic shrinkage of any single concept. The narrative of regime-imposed semantic narrowing applied uniformly to one term is not supported by this measure.

Second, the cPCA-axis non-widening of *liberation* (Section H.5, SD ratio 0.99) is a direction-specific finding rather than wholesale compression. *Liberation*’s total-variance ra-

tio under authoritarianism is 1.046 (slight expansion), and its dominant variance asymmetry favors expansion ($\lambda_{\text{exp}} = 8.25 > \lambda_{\text{comp}} = 6.25$). The SD ratio of 1.00 on the cPCA axis tells us that the algorithm could not find an expansion direction along which Authoritarian *liberation* contexts varied more than Postwar contexts. It does not tell us that liberation discourse globally shrank. The small Postwar sample size ($N = 73$) further qualifies these estimates.

Third, *history* is the term with the most substantial bidirectional compression signal. Its participation ratio drops from 36.5 (Postwar) to 31.7 (Authoritarian), a PR ratio of 0.869, the largest dimensionality compression in the matrix. Despite a positive cPCA centroid shift (+1.85) and a widening cPCA-axis SD ratio (1.22), the term’s contexts collapsed onto noticeably fewer effective dimensions of variation under authoritarianism. The cPCA-axis result (expansion) and the participation-ratio result (dimensional collapse) describe orthogonal aspects of the same change. *History* consolidated its contextual structure while shifting the centroid of that structure along a particular direction.

Under democratization, the picture is uniform expansion. Six of nine terms have trace ratio above 1 under democratization (Democracy, Liberation, and Citizens fall slightly below at 0.983, 0.986, and 0.993 respectively, all within the $[0.95, 1.07]$ band). The same comparison on the participation ratio gives eight of nine above 1, with *unification* (0.969) the lone exception. The Authoritarian-to-Democratic transition is accordingly best read as broad-scale semantic widening on every measure considered.

The bidirectional measures therefore (i) refute the strong reading of authoritarian schooling as a mechanism that compressed liberation discourse specifically; (ii) replace it with a milder finding of cross-term consolidation under authoritarianism, with *history*’s dimensionality collapse as the only large compression signal; and (iii) confirm the broad-expansion reading of democratization on every measure.

H.7 Cluster-bootstrap stability of per-term cPCA shifts

The per-term cPCA shifts in main-text Table 2 are point estimates from book counts ranging from 8 (Postwar) to 22 (Democratic). To test whether the headline pattern of six terms moving with the pooled authoritarian direction, one sitting at zero, and two moving against would replicate in a hypothetical alternative draw of textbooks from the same regimes, we ran a cluster bootstrap by book ($B = 500$ iterations, resampling whole books with replacement within each period so within-book correlation is respected, refitting the cPCA axis on each resample with the same data-dependent α and pooled sign-pinning).

Transition	Term	Full	Median	CI low	CI high	Conc.
Post-Auth	Independence (독립)	+3.26	+2.85	-1.47	+5.00	0.91
Post-Auth	Citizens (국민)	+2.17	+1.83	-1.45	+3.86	0.87
Post-Auth	Nation (민족)	+2.05	+1.83	-0.72	+3.55	0.92
Post-Auth	History (역사)	+1.85	+1.57	-2.98	+4.16	0.84
Post-Auth	Democracy (민주주의)	+0.39	+0.50	-2.47	+3.72	0.61
Post-Auth	Unification (통일)	+0.28	+0.07	-1.30	+1.74	0.54
Post-Auth	Territory (영토)	+0.03	-0.09	-2.04	+2.66	0.44
Post-Auth	State (국가)	-0.69	-0.69	-2.35	+2.55	0.75
Post-Auth	Liberation (해방)	-1.64	-1.32	-3.81	+2.93	0.82
Auth-Dem	Unification (통일)	+2.53	+2.27	-1.79	+4.99	0.78
Auth-Dem	Citizens (국민)	+1.36	+1.28	-1.21	+2.60	0.71
Auth-Dem	Democracy (민주주의)	+0.73	+0.74	-1.21	+2.04	0.75
Auth-Dem	Nation (민족)	+0.65	+0.64	-0.41	+1.86	0.87
Auth-Dem	State (국가)	+0.57	+0.66	-0.33	+1.69	0.88
Auth-Dem	Independence (독립)	+0.13	+0.56	-0.36	+1.47	0.89
Auth-Dem	Liberation (해방)	-0.16	+0.18	-1.25	+1.77	0.41
Auth-Dem	Territory (영토)	-0.71	+0.01	-1.15	+1.57	0.50
Auth-Dem	History (역사)	-1.45	-0.26	-2.49	+3.72	0.56

Table S19: Cluster-bootstrap (by book, $B = 500$) summary for per-term cPCA centroid shifts.

Note: *Full* is the full-data point estimate and matches the cPCA column of main-text Table 2. *Median* and *CI* are the bootstrap median and 95% percentile interval; *Conc.* (sign concordance) is the fraction of iterations whose per-term shift has the same sign as the full-data shift.

We report per-term shifts and 95% percentile bootstrap CIs without multiple-comparisons correction. The 18 simultaneous per-term CIs at nominal $\alpha = 0.05$ imply a family-wise error

rate of approximately 60%, which the appendix-level finding that every per-term CI crosses zero already reflects. Substantive claims in the main text are therefore framed at the pattern level (aggregate signs, cluster structure, the largest-magnitude movements) rather than as per-term significance.

Three observations. First, every per-term 95% bootstrap CI crosses zero. Read as a hypothesis test on whether each term’s cPCA shift differs from zero in a hypothetical alternative draw of textbooks, none of the per-term shifts is individually distinguishable from zero at $\alpha = 0.05$.

Second, sign concordance (the fraction of bootstrap iterations agreeing with the full-data sign) separates the per-term claims by reliability. The four largest-magnitude Postwar-to-Authoritarian shifts (Nation 0.92, Independence 0.91, Citizens 0.87, History 0.84) and the largest-magnitude counter-direction shift (Liberation 0.82) maintain the full-data sign in 82–92% of bootstrap iterations. The smaller shifts (Unification 0.54, Territory 0.44, Democracy 0.61) lie close to chance. Their reported signs in main-text Table 2 are not distinguishable from coin-flips under book-level resampling. Authoritarian-to-Democratic shifts show a similar split, with *liberation* (0.41), *territory* (0.50), and *history* (0.56) at or near chance and *nation* (0.87), *state* (0.88), and *independence* (0.89) substantially above.

Third, the precise full-data partition (seven positive and two against for Postwar-to-Authoritarian, six with and three against for Authoritarian-to-Democratic) is observed in only 20% and 29% of bootstrap iterations respectively. Across iterations, the number of terms moving with the pooled direction is distributed widely (modal range 4–6 for Post–Auth, 6–7 for Auth–Dem). The *existence* of cluster structure (that some terms move with the pooled direction and others against) is preserved in essentially every iteration. The *exact assignment* of terms to clusters is sensitive to which textbooks are sampled.

The implications for substantive interpretation are direct. Claims about the specific signs of small-magnitude shifts (Territory, Democracy, Unification under Postwar-to-Authoritarian; *history*, *territory*, *liberation* under Authoritarian-to-Democratic) cannot be defended at the

level of individual terms. The data are consistent with any of several partitions. Claims about the largest-magnitude shifts (Independence, Nation, Citizens expanding strongly under authoritarian schooling; Liberation moving against the pooled direction; Unification expanding under democratization) survive bootstrap perturbation in at least three-quarters of iterations and are reasonably robust. The 9-term “selective appropriation” tableau as a whole is corpus-specific and should not be read as evidence about a hypothetical larger universe of textbooks.

References

- Abid, A., Zhang, M. J., Bagaria, V. K., & Zou, J. (2018). Exploring patterns enriched in a dataset with contrastive principal component analysis. *Nature Communications*, 9(1), 2134. <https://doi.org/10.1038/s41467-018-04608-8>
- Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of Machine Learning Research*, 3, 993–1022.
- Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794*. <https://doi.org/10.48550/arXiv.2203.05794>
- Hamilton, W. L., Leskovec, J., & Jurafsky, D. (2016, August). Diachronic Word Embeddings Reveal Statistical Laws of Semantic Change. In K. Erk & N. A. Smith (Eds.), *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 1489–1501). Association for Computational Linguistics. <https://doi.org/10.18653/v1/P16-1141>
- Kim, B.-M. (2021). Sentence-embedding-is-all-you-need: Korean sentence embedding repository [Model available at <https://huggingface.co/BM-K/KoSimCSE-roberta>].